

Screening Autism Spectrum Disorder in Children on Facial Images

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By

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Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental disease which has an effect on how individuals connect with others, communicate, grasp. Autism is referred to be a "developmental disorder" since traits develop in the first 2 years of life, despite the fact that any aged people can have it. It is often identified in children. Every individual with autism has unique traits that can vary from mild to severe. People with ASD usually have: trouble in communicating and interacting with people, limited interests and recurring patterns of behavior, social communication etc. Diagnosing ASD in early stages can help patients and doctors in prescribing better treatment. As a result, there is a demand for quick, effortless, and accessible ASD screening methods that can accurately determine whether a patient has ASD based on a set of measured features and advise people whether they should seek a formal clinical diagnosis. However, difficulties persist. Working with datasets that contain data on behavioral features and other aspects like gender, age, ethnicity, etc. is necessary for doing this type of study. There aren't many clinical or screening-related datasets on autism currently accessible. Due to societal, personal, and legal considerations, these data are very delicate and challenging to gather. In this study, we present this problem using Feature Pyramid Network (FPN) to extract the features and use a backbone to predict if the child has ASD or not. The VGG16+FPN model achieved a highest accuracy of 92%.

Abbreviations

ASD	Autism Spectrum Disorder
CNN	Convolutional Neural Network
VGG	Visual Geometry Group
FPN	Feature Pyramid Network
DSM	Diagnostic and Statistical Manual of Mental Disorder
EEG	Electroencephalogram
Fmri	Functional Magnetic Resonance Imaging
ReLU	Rectified Linear Unit

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1. Introduction

Autism Spectrum Disorder (ASD) refers to an intricate brain development disorder. It was first coined in 1911, and the first case of ASD was recorded in 1933. Autism refers to restricted or unvarying activities and faces difficulties with social and communication activities. It may start to develop in infancy and childhood and leads to dis-integrative disorders and Asperger's syndrome. Several symptoms and factors need to be considered to estimate the level of disorder severity because it varies with every individual. So, it is one of the tough jobs to identify a person with ASD.

Generally, the very first symptom of autism spectrum disorder is often noticed during the first year of life, and the children may continue to show signs of autism between 18-24 months of their age which include low communication skills, repetitive behavior, and feeling uncomfortable in a new social setting.

The children may get this disorder because of genetics or surrounding environmental factors which result in affecting the children's social life. They may suddenly become silent or aggressive based on the environmental conditions they are grown up from the past five years. ASD is a chronic disorder that may tend to be noticed during adolescence age as well. So, in simple terms, ASD refers to a neuro-development disorder that affects children's primary forms of communication. In spite of ASD being a chronic illness, many researches and studies proved that early detection and suitable medication and therapies can reduce the level of ASD severity. According to the Diagnostic and Statistical Manual of Mental Disorder (DSM), one in 53 individuals in America is suffering from ASD.

ASD identification is an onerous process because it cannot be diagnosed with a single medical test. Moreover, testing can be various with every individual, so to overcome all these complications doctors are trying to study the childhood development history of that individual to crosscheck any symptoms of autism and conclude the disorder more accurately. There are various steps involved in this procedure. The initial trace of this diagnosis is talking to parents, caretakers, and neighbours of that individual to find out the childhood behaviour and mannerism [4]. Later on, taking all those reviews and behaviour into consideration, the doctor will start taking therapies to minimize the autism severity [10].

There are a few others techniques and, diagnosing procedures will be added to these therapies to know the current status of severity. They are EEG (electroencephalogram) blood tests and fMRI (functional MRI) scans which may contribute to diagnosing the ASD [20]. Autism can be diagnosed with the assistance

of cutting-edge advanced equipment at an early age, say 2 years but, in some cases, it may get delayed up to 4 years of childhood age due to various factors [15].

The main reason for the late diagnosis is the lack of skilled doctors. Additionally, the lack of access to ASD-specific equipment could cause a delay. On the other hand, social and other environmental factors may also cause the delay [8]. We all know autism is a chronic disorder, so people started to think about the early diagnosis which can assist the specialist to identify ASD. There various methods and techniques came into the picture to track ASD.

They are eye-tracking, face recognition, voice recognition, studying the functionality of the brain development and its behaviour, and neurophysiological, and genetic information are the common methodologies to diagnose ASD. After conducting various studies on ASD, they concluded that facial malformations, anomalies, or abnormal facial traits caused during embryonic development are connected to central neurological issues and play a crucial role in diagnosing ASD. In general, doctors try to diagnose ASD with a method called questionnaires, but as the days passed, the accuracy of this technique has been turning down then AIML came into existence

With the advancement of technologies Artificial intelligence (AI) and machine learning (ML) are playing a vital role in ASD Detection. There are various machine learning models like VGG(Visual Geometry Group), CNN(Convolution Neural Network), RESNET(Residual Network) etc. are showcasing the promising result in the medical field. But the main constraint in detecting ASD with the assistance of ASD choosing the appropriate dataset to train our model. During the initial stages, some people started using sensors and a few other physical devices to capture the dynamic EEG and eyeball movements, etc. that can give an edge to identify ASD but they are still under research and scuffling to reach the base accuracy.

So, we are going to replace this inaccurate information with facial images [11]. Numerous studies say that face recognition can produce better results in identifying the person's emotional state, it also assists in extracting the appropriate information while finding out the behavioral patterns [21].

The Machine Learning models will try to extract the facial features from images to classify them into normal and abnormal. Conducting in-depth study to improve the efficiency, sensitivity, specificity, and accuracy of the ASD screening process is a demanding undertaking. Due to societal, personal, and legal considerations, these data are very delicate and challenging to gather. Several studies conducted on facial features extraction produced better results when compared to other diagnosing techniques. The common facial features that ASD person has is wider eyes, substantial mouth, broader face, and short middle face. So, from the past few years the extracted

facial features are playing the vital role for diagnosing the ASD during the early stages. In this study we are going to use Feature Pyramid Network to diagnose the autism with the assistance of convolutional neural network.

2.Related Work

Autism spectrum disorder is a complicated developmental condition that entails persistent issues with social communication, constrained interests, and repetitive behaviour, each individual with autism faces these challenges to varying degrees. It might be challenging to receive a diagnosis of autism; no specific clinical method exists to detect the condition. Kanner was the initial person who identified and published the first methodical illustration of early infantile autism in the year 1943. Only after the scripted work of the Diagnostic and Statistic Manual of Mental Disorders (DSM) published by the American Psychiatric Association [APA][16].

Furthermore, the advanced versions and methods to diagnose autism spectrum disorder were discussed and included in the next revised versions of the DSM. Despite the fact that having many methods to deal with ASD, still inaccurate diagnosis is a potential cause for the substantial growth of the disorder. Accordingly, to reduce the prevalence different methods were brought up along with the digital intelligence incorporating the latest machine learning and deep learning techniques for instance neural networks to analyze and reclassify the inputs given in order for the early prediction. Therefore, according to diverted works, machine learning has made its mark to diagnose autism. A researcher named Lord. et al. have created a method to detect autism-like social communication questionnaires and developed a method to record the essential data through an eye-tracking system using several machine learning classifiers and algorithms[14].

Many other researchers developed different machine learning methods to detect autism spectrum disorder like EyeXplain Autism method[6], supplemental to that an approach that scrutinizes the biomarkers like genetics [3]etc., Mazumdar et al. submitted the work explaining the eye tracking feature classification implemented through machine learning algorithms[17].

As an extension to their work we proposed a model to predict the autism spectrum disorder using a facial image dataset using a machine learning algorithm named Feature Pyramid Network(FPN). Feature pyramid network was first put in place by Mr. Tsung-Yi Lin et al., this is a pyramid structured architecture that helps to detect each and every minute object attained in the image by analysing the given input image stepwise considering the pyramid level as a scale. It is a two-tier architecture using which it detects objects that rest in the image. The basic components of recognition systems for spotting things at various sizes are feature pyramids which help to construct the network. FPN has shown good performance on a satellite imagery dataset in the RGB format for the multi-class land segmentation[22].

FPN solves even a generic problem using a custom backbone in its bottom up layer and it has wide range of choice and exposure and can be paired with different neural network algorithms to solve various problems as discussed in [13]. Zhao et al. replace the top convolution layer with a New Feature Pyramid module as much as possible on the basis of Single-Shot multibox Detection (SSD) for image classification and detection [24].

There are types of Feature pyramid network which are Balanced FPN, BiFPN, CE-FPN each type have its own structured architecture modified from the basic architecture. Currently, we are using the basic FPN architecture for the early prediction of autism spectrum disorder using facial image dataset.

There are few similar works regarding this facial image dataset used to train the machine learning and deep learning model to predict autism spectrum disorder. Hence, in the below-mentioned work, they tried to segregate the facial data as this data is given as an input to the model and would receive four major attributes like facial expressions, morphological features, and emotions that can be noticed in an autism spectrum disorder victim but the accuracy for the approach mentioned in the paper achieved only 76% [12]. Transfer learning is an approach where a pre-trained model is re-used on a new problem, here models like VGG16, ResNet150V2, MobileNetV2 etc., can be used for the transfer learning.

But according to the continuous studies, Feature pyramid network has shown better results when add it to the transfer learning model. Empirical Study of Autism Spectrum Disorder Diagnosis Using Facial Images Dataset by Improved Transfer Learning Approach. This is a research study based out on the transfer learning approach for the early prediction of the autism spectrum disorder. In this paper they worked with VGG19, ResNet50V2, MobileNetV2 and EfficientNetB0 against the facial image dataset were it have achieved better accuracies and one of them have been recognised as a best optimizer which could help the proposed or the working model more efficiently with acquiring the better and improved results reducing the queue time and the true negativity[2].

Methodology

1.1. Convolutional Neural Network

CNN is one of the deep learning methods used to create models for a variety of applications. CNN reduces an image to its lower dimensions without losing its characteristics. Typically, CNN consists of three layers: a convolutional layer, pooling layer and a fully connected layer. [1].

The main layers of CNN are:

- **Convolutional Layer:** It is the first layer used to extract the different features from the input pictures. The convolution operation is a mathematical procedure that is executed at this layer between the input image and a filter of a particular size, $N \times N$. By sliding the filter across the input image, the dot product between the filter and the areas of the image with respect to the filter size ($N \times N$) is calculated as shown in Equation 1. This gives information about the image, including its corners and edges, and is known as the feature map [18].

$$z^n = h^{n-1} * W^n \quad (1)$$

Where z stands for the output layer, h stands for the input to the n th layer and W stands for weight matrix.

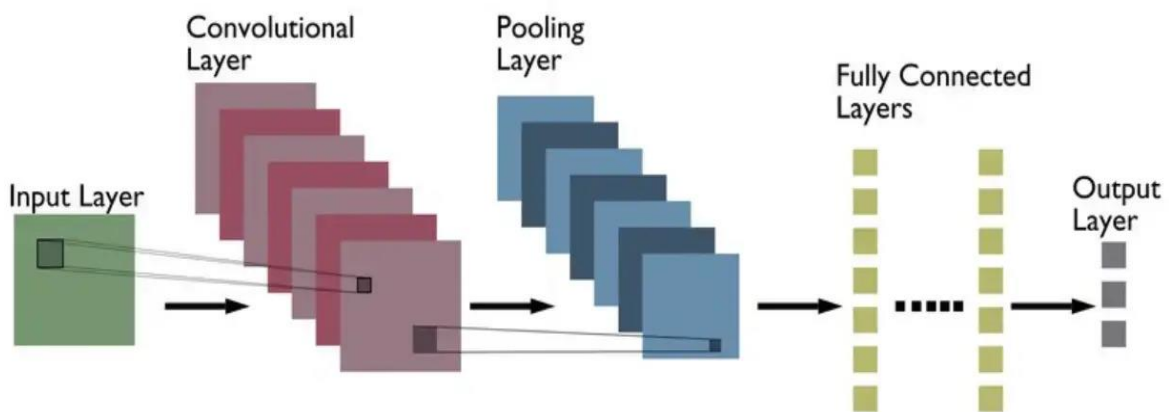


Figure 1: Basic Structure of a CNN Model

- **Pooling Layer:** The spatial dimension of the input is decreased by pooling layers, which makes processing easier and consumes less memory. Pooling

fastens up the training and also helps to a reduction in the number of parameters [7]. The two main types of pooling are max pooling and average pooling. Max pooling uses the largest value, whereas average pooling uses the average value from each feature map. Pooling layers are typically used after convolutional layers to reduce the size of the input before it is sent into a fully connected layer [23].

$$h_{xy} = \max_{i=0 \dots s} \max_{j=0 \dots s} h^{n-1} * (x + i) * (y + j) \quad (2)$$

- **Fully Connected Layer:** One of the fundamental layers of CNN is the fully connected layer. As the name suggests, each neuron in a layer is totally interconnected and coupled to every other neuron in the layer before. The purpose of this layer, which is utilized at the conclusion of CNN, is to combine all the features and findings from the previous layers in order to forecast what will happen in the following layer.

$$z_l = W_l * h_{l-1} \quad (3)$$

In this paper, we implemented CNN on the Autism Face image dataset, here CNN is instantiated with the Sequential Model and the model acquired an accuracy of 85%. The CNN architecture comprises of 7 convolutional layers and 7 max pooling layers.

The input layer shape is (200,200,3) which implies the input images should have a height and width of 200 pixels and 3 color channels (red, green, blue). The used 6 conv2d strides are 16, 32, 64, 128, 256, 512 respectively and the activation function we have used is ReLU with the same padding. This CNN model is defined using the sequential model and it performs a series of convolutional operations followed by max pooling to extract features from an input image. Each convolutional layers have its own set of filters to identify the pattern or features of the input image. The max pooling layers will try to reduce the computation and be less prone to overfitting. At the end of the network, we have added a flattened layer and two dense layers with 512, 1 which gives output as a single value between 0 and 1 of the input images belonging to one of two classes (ASD, Non-ASD).

The architecture of CNN model is shown in Fig. 2.

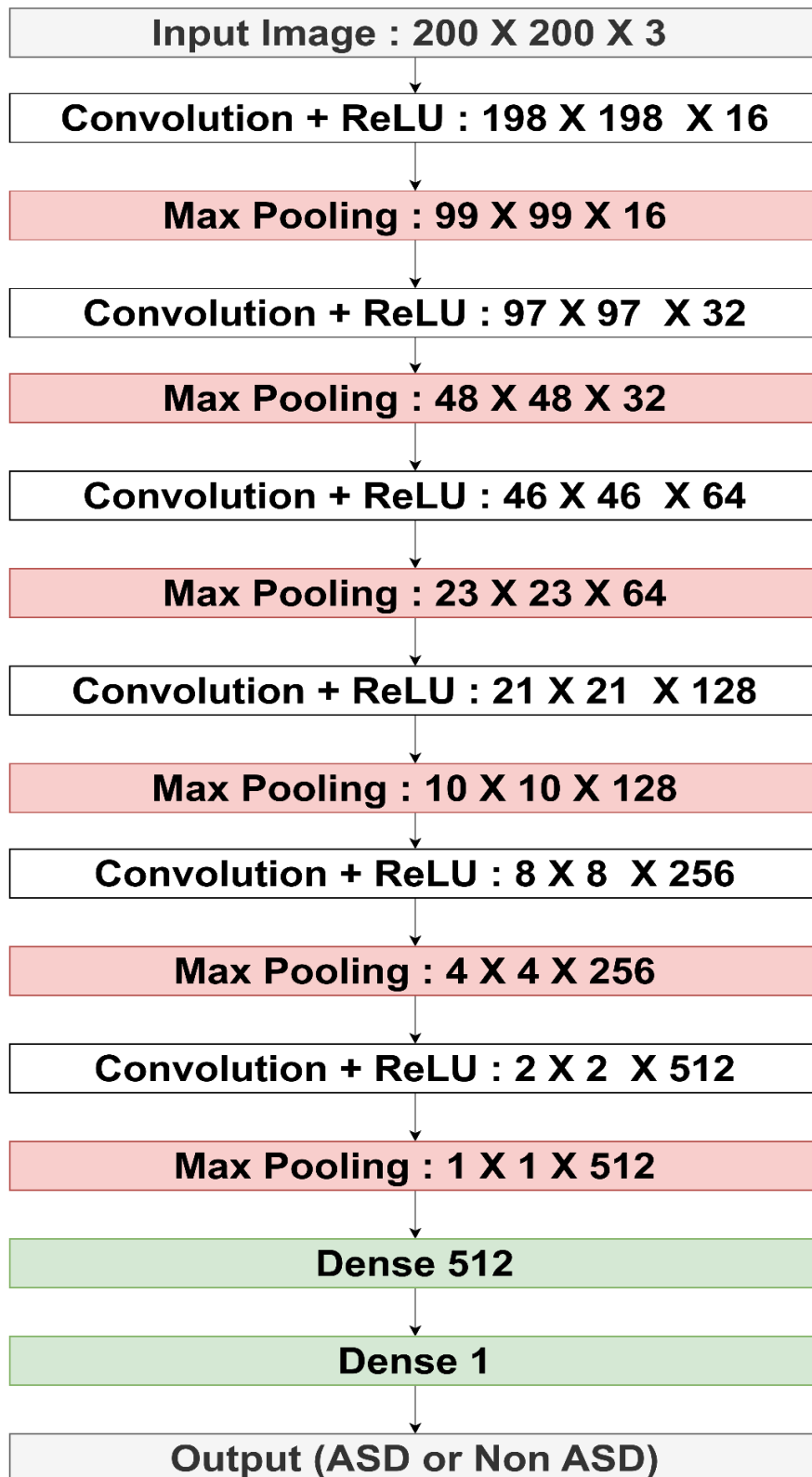


Figure 2: CNN Architecture

1.2. VGG16

Convolution neural networks are among the best vision model architectures available today. VGG16 uses convolution layers with a 3x3 filter, a stride 1, and a maxpool layer of a 2x2 filter of stride 2, as opposed to having a lot of hyper-parameters. In every aspect of the architecture, convolution and max pool layers are placed in an identical sequence [9]. It features 2 fully stratified at the very margin, followed by a SoftMax for output. The 16 in VGG16 denotes the fact that there are 16 layers with weights. There are at most 138 million parameters in this network, appearing to be a large network. However, the VGG16 architecture's simplicity makes the network more appealing [5].

$$Softmax(Z_i) = \frac{e^{Z_i}}{\sum_j (e^{Z_j})} \quad (4)$$

Outline of VGG16 Architecture is as follows:

- **Input:** VGGNet receives an image input of 224×224 . In the ImageNet competition, modelers kept the input image size constant by clipping a 224×224 section from the center of each image [5].
- **Convolutional Layer:** VGG convolutional filters use the smallest possible receptive field of 3×3 . VGG also uses a 1×1 convolution filter as a linear 9 transformation of the input [5].
- **ReLU Activation:** It is the Rectified Linear Unit Activation function. ReLU is a linear function that produces zero when the input is negative and an equivalent output when the input is positive. Because VGG uses a 1-pixel convolution stride, the spatial resolution is maintained after convolution [5].

$$ReLU(Z_j) = \max(0, Z_j) \quad (5)$$

- **Hidden Layer:** All hidden layers of a VGG network use ReLU instead of regional response normalization such as AlexNet. The latter increases the training and memory consumption, but the overall accuracy does not improve much [5].
- **Pooling Layer:** The Pooling layer accompanies multiple convolution layers - this helps to minimize the dimensions and number of parameters of the feature maps produced by each convolution step. Because the number of available filters soon rises from 64 to 128, 256, and 512 in the final levels, pooling is essential.

- **Fully Connected Layer:** Three completely interconnected layers make up VGGNet. Each of the first 2 layers contains 4096 channels, while the third layer contains 1000 channels – one for each class.

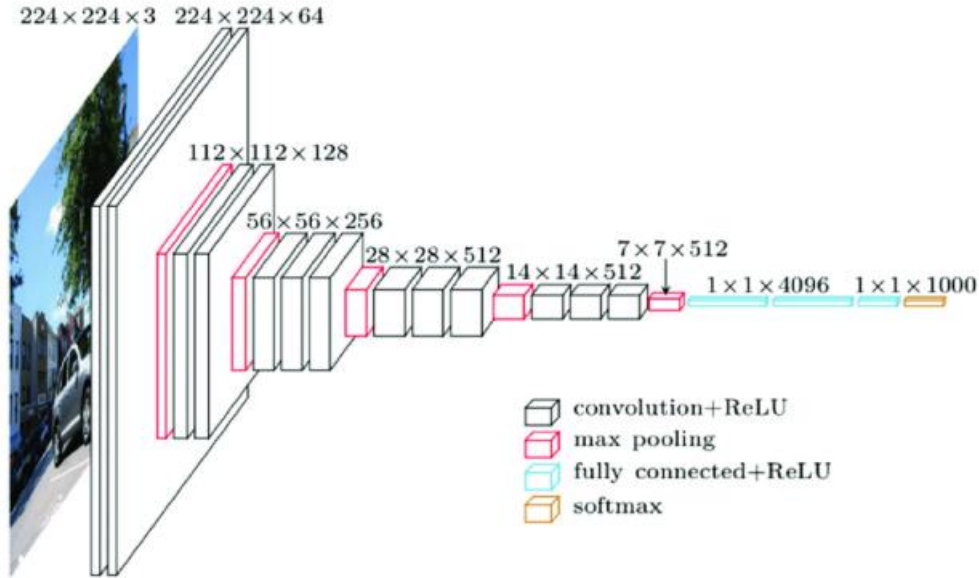


Figure 3: VGG16 Architecture

In this paper, we implemented VGG16 on Autism Face image dataset, pre trained VGG16 model is used and the model acquired an accuracy of 89%. The VGG architecture takes the input size as $(224, 224, 3)$. The 5 conv2d strides are 64, 128, 256, 512, and 1024 respectively and the activation function we have used is ReLU. Then we have defined the convolutional layers, pooling, fully connected and Dense layers.

1.3. Feature Pyramid Networks (FPN)

Before deep diving into the core concept, the main idea behind incorporating this feature pyramid network into the deep learning model is to improve the feature extraction from the given data, so that the model performance may get refined. It is completely a divergent approach that focuses on generating multiple feature maps with various resolutions. The input image will get extracted layer by layer with different resolutions and passes the outputs of the previous layer to the succeeding layers [13]. Generally, FPN is used in object detection because it produces feature maps with various resolutions which resolved one of the most arduous problems of identifying the objects in the input image of different sizes. Conventional

architectures like CNN, ResNet, VGG, etc. will produce only a single output feature at the end of the network.

So, in most cases, the conventional convolutional neural network models fail to discern the smaller objects present in the image. Sometimes these feature maps may not be able to detect large objects as well [22]. To accommodate this concern Feature Pyramid Network came into the picture that provides multiple features with different resolutions at various layers, which leads to the formation feature pyramid. This entire process will be taken place in two consecutive phases.

The first phase is independent of any neural network often we call this phase the backbone of the network and the second phase is the construction of our pyramid which combines outputs from two different layers and produces the up-sampled feature map. These two are connected by lateral layers. As discussed above the Feature Pyramid Network architecture associated with a bottom-up pathway also known as backbone, a top-down pathway, and a lateral connection between these two will make a complete FPN network [13].

The bottom-up pathway can be a feed-forward neural network that can produce feature maps. In this paper, we have used two networks they are Convolution Neural Network, and the other one is VGG16 (Transfer Learning). In CNN we have used the last three layers as the backbone for this FPN with a scaling factor of 2. In general, all these layers will be considered as a single stage but in FPN we have added a lateral connection and passed the old feature map output as a reference to the corresponding top-down layer which enhances the integrity of feature maps.

The first layer of this backbone has more features when compares to the last layer. At every layer, we will try to store the output. In this particular CNN architecture, we have used the last three residual blocks and named them C3, C4, and C5. These layers have strides of 128,256,512 with respect to the input image of size (224,224). We have not used the first three layers as the backbone because of its large memory.

The backbone will produce the low-resolution image of the input with more semantic value and this feature map will be stored in C5 parallelly we will store all the individual layer outputs as well at each layer. Now moving on to the second phase the top-down pathway takes the C5 output from the

backbone as an input image feature which has a high semantic value from a higher pyramid level and passes on to the next layer. At every top-down layer, there will be a lateral connection from the bottom-up pathway at each layer.

The lateral connection performs a 1×1 convolution operation to make sure all the images at every layer is having the same output shape. Then we tried to add the element-wise later on we applied a 3×3 convolution to all the new integrated layers. This process continues till it reaches the end of the top-down pathway, which provides three different output feature maps of different resolutions named P3, P4, and P5.

Let us illustrate with an image

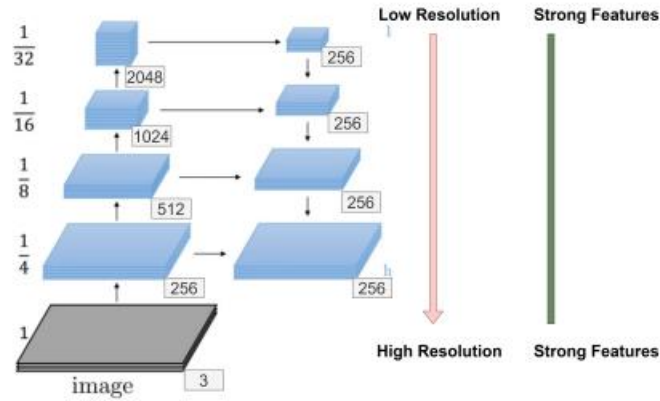


Figure 4: FPN Architecture [13]

In Figure 4 you can observe that the first phase is a bottom-up pathway constructed using a convolutional network for extracting the features. As we move towards the top the image resolution will get decreased and the semantic value for each layer will increase consistently. Now the final output of the backbone which consists of a strong semantic value feature map is now passed on to the top-down pathway layer.

But the main concern here the restoring layers have higher semantic values but the objects' locations are still imprecise. So, the lateral connections will play their role to improve the feature map by adding the outputs of the previous layer to the current feature map. By the end of the network, we have three output feature maps named P3, P4, and P5 corresponding to C3, C4, and C5 of the same spatial size.

Here we have used nearest neighbour up-sampling for simplicity and we set depth $d = 256$. This filter will reduce the blurring effect when merging the two feature maps in the up-sampling layer. We can use any one of the feature maps to classify the output based on the dataset. We have used three feature maps and recorded the accuracies of all the feature maps.

1.4. CNN + FPN

We have used the general convolution neural network as backbone for FPN. The CNN architecture comprises of 7 convolutional layers and 7 max pooling layers. The input layer shape is (200,200,3) which means the input images should have width and height of 200 pixels and 3 color channels (red, green, blue). The 6 conv2d strides are 16, 32, 64, 128, 256, 512 respectively and the activation function we have used is Relu with same padding.

This backbone model is defined using the sequential model and it perform series of convolutional operations followed by max pooling to extract the features from the input images. Each convolutional layers have its own set of filters to identify the pattern or features of the input image. The max pooling layers will try to reduce the computation and less prone to overfitting.

The bottom-up pathway is built now we have our lateral connections and outputs C3, C4, and C5 at every intermediate layer. Now we have to pass the C5 output to the top-down pathway. The top-down pathway consists of three layers P3, P4, and P5 which are up-sampled and combined with the feature maps from the backbone of low resolutions C3, C4, and C5 respectively. The up-sampling action is performed using the bi linear interpolation technique, which maintains the spatial information and gives a smooth output.

The first 1x1 convolution operation is carried out on highest scale feature map P5 to reduce the dimensionality. The next highest scale feature map P4 is up-sampled by performing the merging operation with P5 in element wise manner. Similar process will be repeated with P3 as well. Once we get the feature maps P3, P4, and P5 we perform the flatten operation and passed it onto the 512-neuron dense layer with Relu as an activation function. Later on, the output layer is then defined as a dense layer with a sigmoid activation function. The output layer will be used as the classification head and produces a single output value between 0 and 1, which denotes the prediction probability for the input image belonging to that particular class.

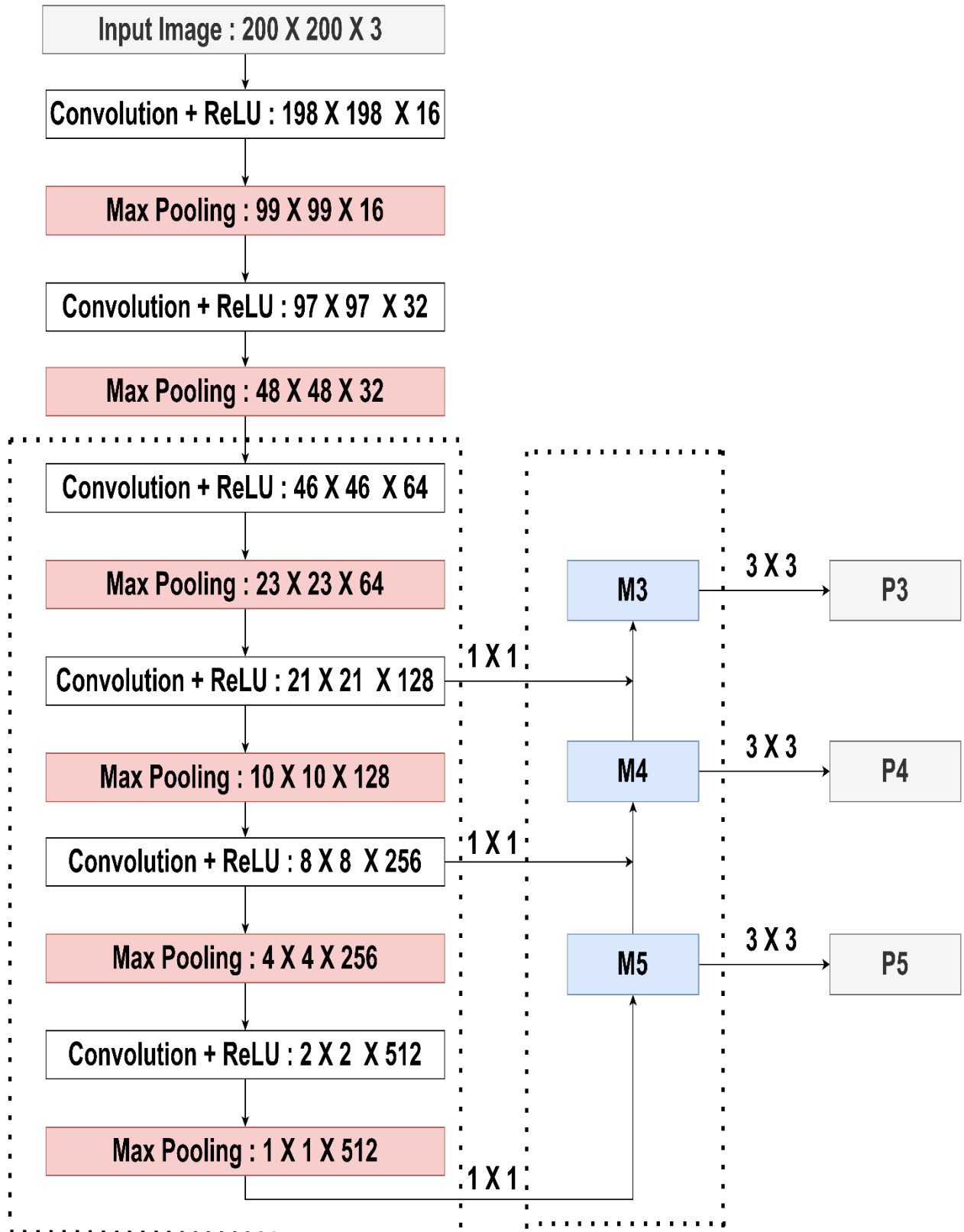


Figure 5: CNN + FPN Architecture

1.5. VGG16 + FPN (Transfer Learning)

VGG16 (Visual Geometry Group 16) is a deep learning convolutional neural network architecture which was first introduced in 2014. It is a prominent network for image classification and has accomplished state-of-the-art results on various benchmark datasets. Feature Pyramid Network (FPN) is a methodology used in object detection tasks to identify objects at various scales in an image, segmentation and other computer vision tasks.

Using a pre-trained model as initial foundation for a contemporary task could be better option rather than building a new model from scratch is a machine learning approach known as transfer learning. The pre-trained model is often trained on a big and diverse dataset, and even if the new dataset is smaller or different from the original dataset, the knowledge obtained from this training can be applied to the new task.

VGG16 is often used as a pre-trained model for transfer learning in object detection tasks. FPN can be added on top of the VGG16 convolutional base to detect objects at multiple scales in an image. Here we used the VGG16 layer network as the backbone for FPN. For doing this, first, we need to load the weights of the pretrained VGG16 model.

Here, we initialize the weights to “imagenet” to utilize the pre-trained weights from the ImageNet dataset, Here the final layer is set to False to remove the fullyconnected layer at the peak of the neural network, and input-shape to (224, 224, 3) to resize the input images to that size. The input layer shape is (224,224,3) which refers to the input images where they have a width and height of 224 pixels and 3 color channels (red, green, blue). The 5 conv2d strides are 64, 128, 256, 512, and 1024 respectively and the activation function we have used is Relu.

Now, we define the top-down pathway of FPN. The three convolutional feature maps C3, C4, and C5 are extracted from the VGG16 model and are used to generate the feature pyramid network. The top-down pathway consists of three layers P3, P4, and P5 which are up-sampled and combined with the feature maps from the backbone of low resolutions C3, C4, and C5 respectively. The final feature pyramid output is stored in the variable P3.

So, let us consider the architecture of the model.

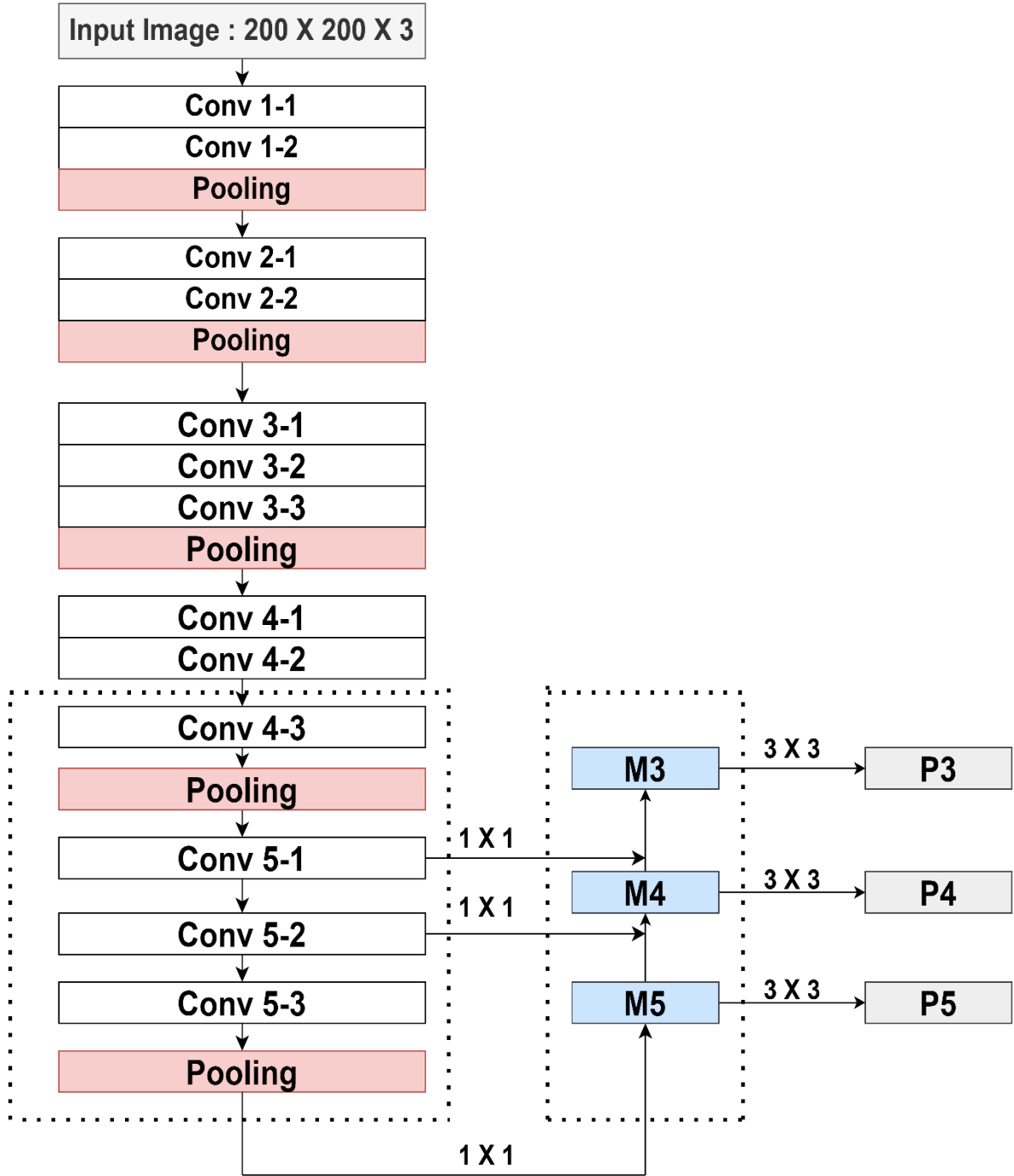


Figure 6: VGG16 + FPN Architecture

Once the feature maps are obtained, we then perform flatten operation. The Flatten layer is used to convert the 3D tensor output of the FPN model into a 1D tensor that can be fed into the fully connected layers. Then we have defined several Dense layers with different numbers of units and different dropout rates. Each Dense layer is followed by a ReLU activation function that

introduces non-linearity into the model. Dropout layers are used to prevent overfitting in the model. Finally, we have a single Dense layer with a sigmoid activation function that outputs a binary classification prediction indicating the probability of the input image being of a particular class.

Results

1.6. Dataset Description

It is an open-source data named Autism Facial Images which is available in Kaggle. This dataset consisted of facial images of children who are Autistic and Non-Autistic. The data set is appropriate for classifier training. In the training set, there are 1327 visuals of children who are autistic and 1327 images of children who are not autistic. The test set includes 140 facial images of children who are autistic and 140 images of children who are not autistic. Some of the images from the dataset are shown in Figure 7.



Figure 7: Sample Images from Dataset

1.7. Evaluation Metrics:

In this paper, we have used various metrics to evaluate the model performance they are accuracy, sensitivity, and specificity. The sensitivity of the model will be evaluated whether the deep learning model perfectly recognizes the images of the faces who are suffering from autism ("True Positives").

The sensitivity metric is used to find out how frequently a positive prediction of autism spectrum disorder was right. It was calculated using the Formula 6.

$$Sensitivity = \frac{True\ Positive}{True\ Positive + False\ Positive} * 100 \quad (6)$$

Specificity deals with non-autistic (“True Negative”) images of the children's faces. It was used to find out how frequently a negative prediction is right. It was calculated using the Equation 2 and accuracy is the degree of closeness between the predicted value to its true value, the total accuracy of the model is evaluated using the Equation 3.

$$Specificity = \frac{True\ Negative}{True\ Negative + False\ Negative} * 100 \quad (7)$$

$$Accuracy = \frac{TP+TN}{FP+FN+TP+TN} * 100 \quad (8)$$

1.8. Quantitative and Qualitative Results

The deep learning models which were proposed in the paper were trained using the Google Colab Environment. Every model was trained for 100 epochs with 8 as batch size. The results of Basic Convolutional Neural Network, VGG, CNN with FPN and VGG with FPN are noted down in Table 1.

<i>Model</i>	Testing Accuracy	Training Accuracy
CNN	85	100
CNN + FPN	88	100
VGG16	89	99.89
VGG16 + FPN	92	99.36

Table 1. Results

The above results of the training and testing accuracies of four distinct models: CNN, CNN+FPN, VGG, and VGG+FPN. The CNN model is having higher testing accuracy of 100% but the testing accuracy of the model is only 85% which shows that the model may be overfitted with the training data, and denotes the model is performing good on the training data but when passing the new data as input to the model it is not upto the mark.

So, we tried to tune this model performance with Feature Pyramid Network. The CNN+FPN model has obtained a training accuracy of 100 and a testing accuracy of 88. Adding the FPN to the CNN model has improved the testing accuracy by 3%. The Feature Pyramid Network has managed to extract the features at different resolutions which may have assisted the model to get more information from the input images.

The VGG model has obtained a training accuracy of 99.89 and a testing accuracy of 89. The VGG model has additional layers when compared to the CNN model which assisted to grasp additional features from the input image. The VGG model has managed to generalize the new data but we tried to add FPN to the VGG to check for any improvement in the model performance. The addition of FPN has increased the model testing accuracy by 3% from the previous best. The training accuracy of the model is 99.36 and the testing accuracy of 92.

In conclusion, The VGG+ FPN has the highest accuracy followed by the VGG model. The addition of FPN to both the CNN and VGG models has noticed an improvement in their performance which showcases the significance of feature extraction in deep-learning models.

The Autism Face Image dataset is diverged into a training set and testing set in the ratio 80:20 on which the 4 models are applied. In this work, every model is trained for 100 epochs and we have used various metrics to evaluate the performance. And for evaluation we have used several measures. The applied measures are Precision, Recall, F1-Score, ROC Curves as shown in Table 2.

<i>Model</i>	Precision	Recall	F1-Score
CNN	0.88	0.81	0.84
CNN + FPN	0.84	0.92	0.88
VGG16	0.86	0.92	0.88
VGG16 + FPN	0.91	0.93	0.92

Table 2. Performance Metrics

Classification Report and Confusion matrix are also used to calculate the sensitivity, specificity and accuracy. Sensitivity and specificity are calculated in Table 3.

<i>Model</i>	Specificity	Sensitivity
CNN	0.71	0.89
CNN + FPN	0.92	0.82
VGG16	0.92	0.85
VGG16 + FPN	0.95	0.86

Table 3. Performance Metrics

More insights can be obtained by plotting the accuracy, loss plots and ROC curves. The accuracies, losses and ROC curves are recorded in Figure 8, Figure 9 and Figure 10 respectively. The ROC (Receiver Operating Characteristic Curve) shows us the performance of a classification model at all thresholds. This is a curve between true positive rate and false positive rate. By lowering the classification threshold, more items are classified as positive, which raises the number of both False Positives and True Positives.

$$TPR = \frac{TP}{TP+FN} * 100 \quad (9)$$

$$FPR = \frac{FP}{FP+TN} * 100 \quad (10)$$

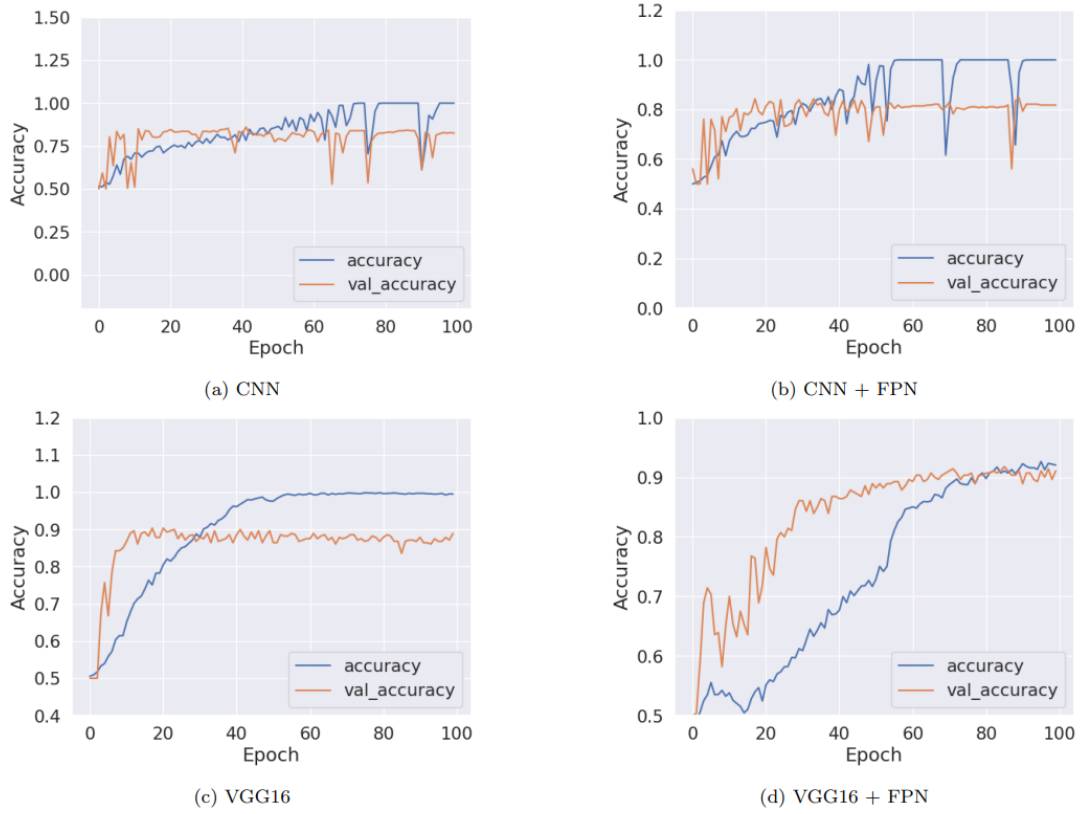


Figure 8: Accuracy Plots

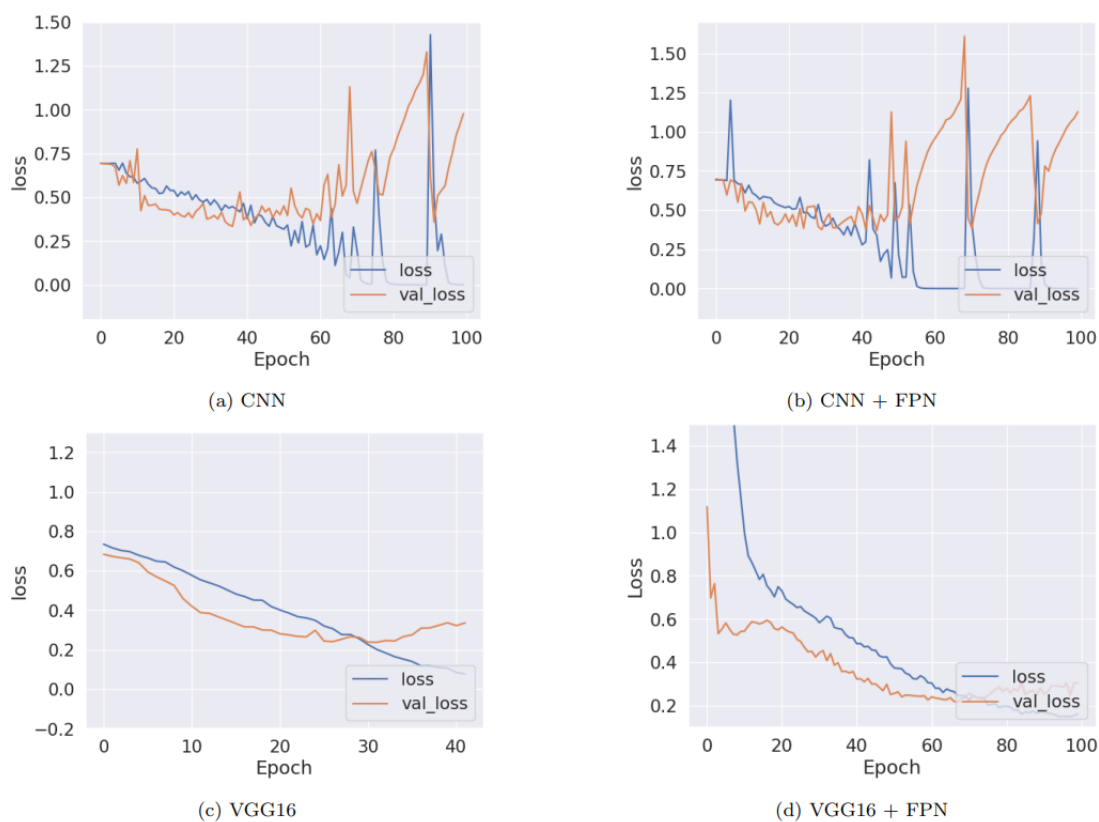


Figure 9: Loss Plots

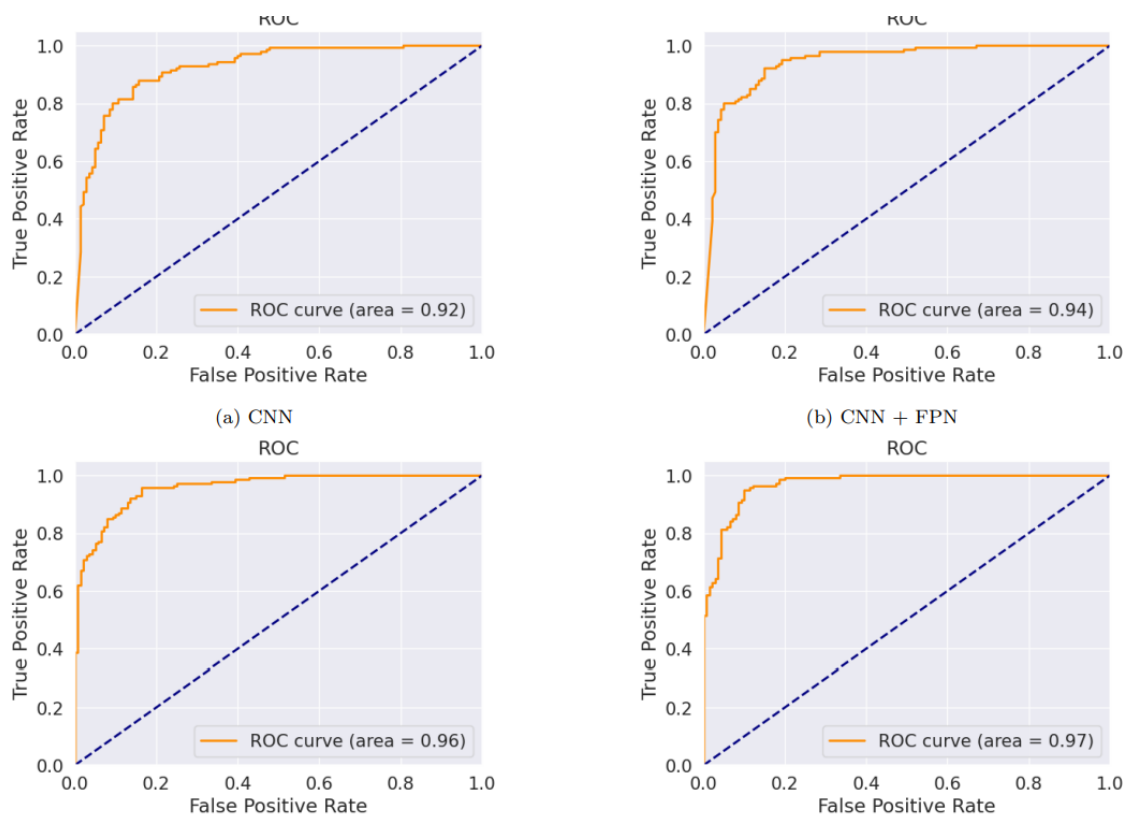


Figure 10: ROC Curves

Conclusion and Future Work

In recent years deep learning acquired a major position in healthcare sector. We used a open source dataset which is available in Kaggle. Our work focused on identifying whether a child is diagnosed with ASD or not. We have considered CNN, VGG16, CNN+FPN and VGG16+FPN models for our work. Amongst the four models, VGG16+FPN, a transfer learning based approached achieved highest accuracy of 92\%. The addition of FPN to CNN and VGG16 has improved the performance which demonstrates the importance of feature extraction in deep-learning models. We extend our work as seeing a future scope considering the advancement of technology and many ongoing research works. We can consider ECG, Speech recognition, behavioural, and iris data alongside the facial image dataset to acquire better results with improved accuracy for the early prediction of the autism spectrum disorder within infants to reduce the prevalence.

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